

Spatio-Temporal Anomaly Detection for 5G-Clusters: A Multi-Scale Fuzzy Contrastive Learning Approach

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Abstract—Anomaly detection is an important data-mining task closely related to specific applications. In recent years, spatio-temporal data deriving from various networks has been collected from many real-world scenarios, but how to detect anomalies in such data is still an open problem. In this article, we propose a novel anomaly detection framework, Multi-scale Fuzzy Contrastive Anomaly Detection (MFCAD), by capturing anomalous patterns of data from multiple spatio-temporal scales so as to learn distinguishable feature representations. Different from conventional reconstruction- or prediction-based anomaly detection methods, this approach is not concerned with the consistency of the encoded representation of the implicit layer and the discriminability of the implicit layer of the anomalies. MFCAD implements a more discriminative representation using fuzzy contrastive learning and explicitly performs anomaly detection in the potential space by measuring the distance between outliers and implicit values. This proposed method has been applied for the anomaly detection of 5G mobile network clusters (5G-MNCs) in China Mobile. Furthermore, in order to validate the generalizability of the proposed method, it is further tested on public datasets and experiment results show the promising performance.

Index Terms—Anomaly detection, fuzzy contrastive learning, graph learning, spatio-temporal modeling.

I. INTRODUCTION

ANOMALY detection aims to discover anomalous points/patterns from data that do not match normal patterns. Anomaly detection plays a key role in many real-world applications, such as network intrusion detection [1], financial fraud detection [2], equipment failure detection [3], health monitoring [4], and so on. Mobile networks are the current infrastructure for wireless communication. In recent years, 5G mobile networks are gradually replacing 3G/4G networks. The millimeter wave band of 5G technology brings

greater bandwidth and low latency, but it also leads to a significant increase in the number of wireless base stations. By the end of June 2023, China has more than 2,937,000 5G base stations for commercial operation and 676 million 5G users. Facing such a huge number of base stations, how to effectively and timely detect anomalies has become a key issue in daily operations.

5G mobile network clusters (5G-MNCs) can be defined as collaborative regional systems within a specific coverage area formed by a group of highly correlated base stations. For instance, an example of 5G-MNC is shown in Fig. 1, where each node represents a base station, and the edge represents the correlation between two base stations. It can be observed that even within the same cluster, the degree of mutual influence between base stations varies due to geographical distance. Maintaining the correlation between wireless base stations is essential for ensuring continuous coverage and efficient load distribution, facilitating signal switching and load balancing as users move. To achieve these goals, collaboration and virtualization techniques are employed to enable resource sharing and dynamic scheduling within clusters. As a result, data generated by 5G-MNCs exhibits inherent both spatial and temporal dependencies [5]. Due to the aforementioned spatio-temporal characteristics, it significantly increases the complexity of anomaly detection for 5G-MNC data.

In the field of anomaly detection in mobile networks, conventional approaches manually set fixed thresholds for Key Performance Indicators (KPIs) of different base stations, which requires the experience of domain experts [6]. Furthermore, these methods are designed for univariate sequences, which means that each base station is monitored individually. However, multiple base stations in 5G-MNCs interact with each other due to the cooperation in a specific coverage area. Consequently, it is unable to capture the spatio-temporal synchronization correlations existing in the 5G-MNC data and cannot reflect the overall state of the system well [5].

Recently, the efficacy of deep learning-based techniques for anomaly detection has been well-established [7], [8], [9]. Particularly, Thill et al. [10] introduced an auto-encoder (AE) model based on temporal convolution, effectively applying it to anomaly detection in electrocardiogram (EEG) recordings. Li et al. [11] devised a superimposed variational autoencoder (VAE) model employing graphical neural networks (GNNs) for time series anomaly detection. Deng and Hooi [12] proposed a predictive model based on GNN,

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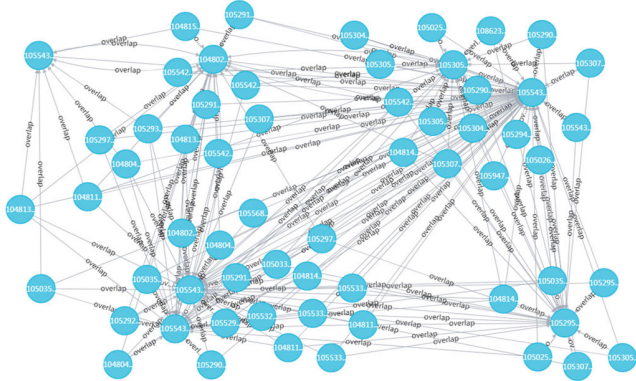


Fig. 1. An example of 5G-MNCs.

leveraging the discrepancy between predicted and actual values to detect anomalies. He and Zhao [13] harnessed Temporal Convolutional Networks (TCNs) for anomaly detection, utilizing prediction errors to compute anomaly scores. These methodologies primarily ascertain whether a sample is abnormal by evaluating reconstruction or prediction errors, since they assumed the model would fail to reconstruct or forecast the unseen abnormal samples. Nonetheless, their focus primarily rested on reconstruction or prediction performance, potentially overlooking the distinction between representations of anomalous and normal samples. As a result, even anomalous data might be well reconstructed or fitted [14]. Recently, by possessing the ability to obtain discriminative features, contrastive learning was also applied in temporal data anomaly detection [15], [16]. However, existing works did not take into account spatial factors and the multi-scale characteristics of data in real applications. Furthermore, these approaches did not readily apply to anomaly detection within 5G-MNC environments, as commercial 5G networks often exhibit intricate dynamic temporal-spatial traits.

In this study, we have discovered that anomaly detection can be effectively carried out within the latent space by measuring the representation distances between outliers and normal data. Notably, this approach is achieved without reliance on reconstruction or prediction models. Because the latent representations are relatively low-dimensional compared with the original 5G-MNC data with high dimensionality and complicated structure. Performing anomaly detection on the compact latent representations helps to concentrate solely on the features that distinguish normal and anomalous. Based on this idea, we propose a novel framework named MFCAD, designed to tackle the challenge of anomaly detection in 5G-MNCs. Firstly, we elaborate on an effective data augmentation method for specific KPI time series in 5G-MNCs, where fuzzy mechanism is involved for the nonlinear data processing. And, a multi-scale fuzzy contrastive module is designed to extract temporal and spatial features from raw data. To capture anomalous behaviors at different scales, our proposed framework employs two GCN-based models that simultaneously conduct node-level and subgraph-level contrastive learning. Subsequently, the data is projected from the temporal and spatial modules into distinct representations within the latent space. Considering that KPI time series are often influenced

by external factors and exhibit subtle pattern variations across different periods, fuzzy C-means (FCM) [17] is applied to cluster the learned representations from various time steps. This process aims to derive corresponding prototypes. Finally, the distances between the representations and their corresponding prototypes are utilized for anomaly inference, thereby enhancing the robustness of detection performance.

The main contributions of this work are summarized as follows:

- A novel framework for spatio-temporal anomaly detection is proposed, which not only is notably suitable for 5G mobile networks in real-world scenario, but also exhibits good performance on other public datasets.
- A multi-scale contrastive learning is first proposed to capture anomalous patterns by obtaining temporal and spatial features at different scales, where differences in normal and abnormal features at diverse data granularity are utilized.
- To better generalize the generated positive and negative samples, intuitionistic fuzzy information is involved in data augmentation, which combines with multivariate dynamic time warping (MDTW) to achieve robustly metric among samples.
- The real 5G-MNC datasets and two sensor datasets are used for validation. Experimental results demonstrate the capability of the method to achieve anomaly detection in real-world scenarios.

II. RELATED WORK

A. Anomaly Detection in Multivariate Time Series

In many scenarios, spatio-temporal data is presented in the form of multidimensional time series. Existing researches on anomaly detection in multivariate time series can generally be divided into two categories: prediction-based [12], [13] and reconstruction-based methods [10], [18], [19], [20]. Prediction-based models detect anomalies by comparing the deviation between predicted values and actual values. For instance, Deng and Hooi [12] proposed a method that combines a graph structural learning module with Graph Neural Networks (GNN) to detect anomalies based on the deviation between predicted and actual values. He and Zhao [13] applied a temporal convolutional network for anomaly detection, where the prediction errors are fitted into a multivariate Gaussian distribution and used to compute anomaly scores.

Reconstruction-based models typically utilize auto-encoders (AE) or variational auto-encoders (VAE) [10], [19], [20] to learn representations from normal data and perform anomaly detection based on the reconstruction error between real samples and reconstructed samples. Thill et al. [10] proposed a temporal AE architecture for anomaly detection, where the reconstruction error is used as an indicator of anomalous data. Su et al. [19] introduced a recursive neural network for anomaly detection in multivariate time series data, utilizing reconstruction probability to identify anomalies. Xu et al. [6] proposed an unsupervised algorithm based on a variational auto-encoder for anomaly detection in seasonal KPI data. Zong et al. [20] designed a deep auto-encoder and Gaussian

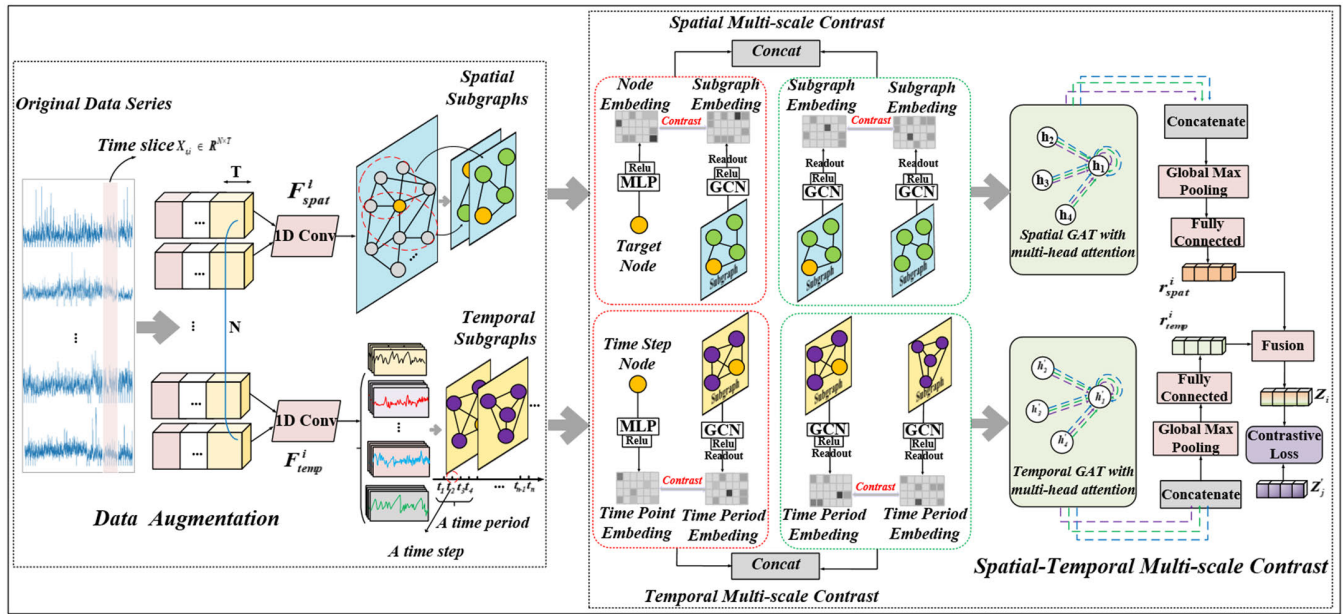


Fig. 2. Framework of the proposed MFCAD.

mixture model called DAGMM for modeling the density distribution of multi-dimensional data. However, these methods typically focused solely on anomalies in either the temporal or spatial aspect. As data with spatio-temporal features often exhibits interdependencies and mutual influences, previous methods may overlook underlying influences.

Recently, temporal and spatial dimensions in multivariate time series have been explored [21], [22], [23], [24], [25], [26]. For instance, Tian et al. [21] introduced an anomaly detection network that incorporated temporal and spatial information, aiming to enhance the differentiation between normal and anomalous instances by reconstructing the prediction error. STDTB-AD [22] extracted features from trajectory data by computing motion statistics within spatio-temporal units to handle the heterogeneity of behaviors. Hu et al. [24] proposed an anomaly detection algorithm based on the fusion of target spatio-temporal context to fully leverage the spatio-temporal contextual information of the targets for anomaly detection. 3G-Net [25], an end-to-end deep learning framework, was proposed for detecting anomalies in spatio-temporal data in the Internet of Things. Wang et al. [26] proposed a framework for anomaly detection based on spatio-temporal graph neural network, utilizing spatio-temporal graph neural network technique to extract feature data from information graphs containing spatial and temporal features.

Both time and space can be divided into different scales. For example, the difference in sampling frequency on the temporal dimension leads to differences in the details of data fluctuation features. And, the differences in the scope of data aggregation at the spatial scale will affect the learning of relevant features of adjacent data. However, the existing methods commonly did not consider the impact on multiple spatio-temporal scales. And, most methods also relied on the reconstruction or prediction. Here, we incorporate multi-scale spatio-temporal features into contrastive learning and perform

the anomaly detection in a compact latent space without relying on reconstruction or prediction models.

B. Multi-Scale Contrastive Learning

Contrastive learning is a self-supervised approach. It has achieved significant performance in the fields of computer vision (CV) [27] and natural language processing (NLP) [28]. The objective of contrastive learning is to learn better feature representations of unlabeled data through unsupervised learning, maximizing the consistency between samples and their augmentations, thereby facilitating their application in downstream tasks [29]. Many researchers have explored the application of contrastive learning in representation learning for time series data [30], [31], [32]. Tonkaboni et al. [30] proposed a self-supervised framework for learning a generalized representation of non-stationary time series. Eldele et al. [31] designed an unsupervised learning framework that leverages time and context for contrastive learning, learning representations of time series. Zhong et al. [32] applied contrastive learning to clustering to increase inter-class differences and reduce intra-class differences. However, these methods overlook the complex relationships among spatio-temporal data.

In recent years, researchers in the field of graph anomaly detection have adopted multi-scale contrastive methods [33], [34], [35], [36], [37]. For example, Jin et al. [33] proposed ANEMONE, a framework based on multi-scale contrastive learning that captures richer anomaly information through node-level and context-level contrasts. Zheng et al. [34] introduced a self-supervised learning framework called SL-GAD. This approach utilized multi-scale contrastive methods to construct different contextual views, allowing for the capture of anomalies in structural space and enabling graph anomaly detection. Zhang et al. [35] integrated multi-view contrastive learning and attribute reconstruction-based models to address anomaly detection in attribute networks. However,

these existing methods did not fully exploit the intrinsic relationships among spatio-temporal data.

Recently, temporal data anomaly detection employing contrastive learning is attracting the attention of researchers. For instance, in 2023, MGCLAD [15] adopted contrastive learning to capture multiple dependencies of signals to enhance the discriminative power of feature representations. Meanwhile, DCdetector [16] introduced a dual-attention asymmetric design that utilized contrastive learning to learn features obtaining an invariant representation. Both of the above studies achieved relatively competitive performance in time series anomaly detection by utilizing contrastive learning. However, there are still some open problems. Firstly, existing works mainly focused on temporal data. But, as to spatial-temporal data, more complex correlations and mutual influence existing in both temporal and spatial dimensions. In this case, how can one implement anomaly detection applying contrastive learning to obtain discriminative feature learning? Secondly, both temporal and spatial dimensions essentially have multi-scale characteristics, and obtaining various features from different scales is meaningful. But existing works still extracted features from temporal data based on a single scale, often neglecting the learning of multi-scale spatial features.

Motivated by this insight, this study transforms spatio-temporal data into corresponding graph structures and applies multi-scale contrastive methods from both temporal and spatial perspectives to capture more comprehensive anomaly information. Through contrastive learning, the learned latent representations can effectively distinguish outliers from normal values.

III. METHOD

The proposed MFCAD framework is illustrated in Fig. 2. The data augmentation module is used to generate enhanced positive and negative samples for subsequent feature learning from the raw 5G-MNC data. Subsequently, the spatio-temporal data is transformed into corresponding graph structures. Correspondingly, multi-scale contrastive learning is then performed from temporal and spatial perspectives. And, two graph attention networks are involved, as GAT has the inherent capacity for relational modeling and the contrastive loss ensures the discriminability of representations in the latent space. The following sections provide more detailed information.

A. Problem Formulation

In this article, we transform the problem into an anomaly detection task for spatio-temporal data with multiple variables. The anomalous behaviors of neighboring nodes are to be captured by converting the spatio-temporal features into graph structures. Each univariate sequence represents a specific Key Performance Indicator (KPI) sequence generated by a base station in 5G-MNCs. The KPI data in mobile networks typically exhibit temporal correlations within each dimension sequence (i.e., periodicity of sequences generated by a particular base station) and spatial correlations between different dimension sequences (i.e., collaborative relationships among base stations in a specific coverage area). We consider a dataset Ω

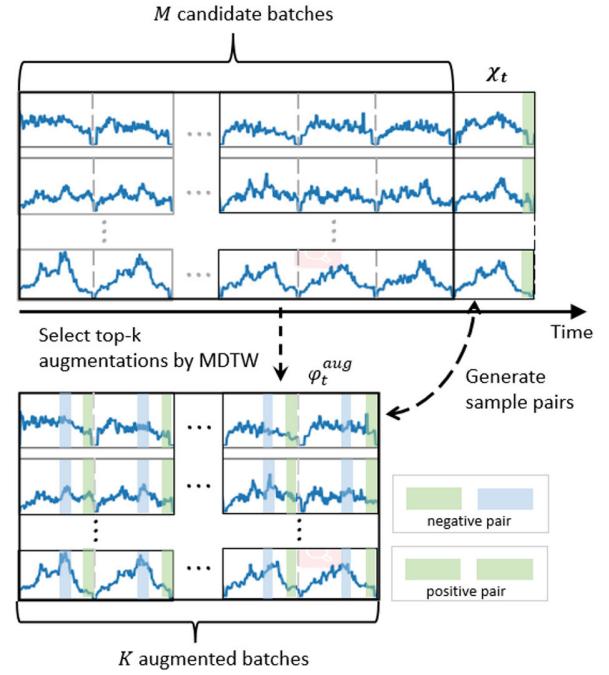


Fig. 3. Framework of the designed data augmentation method.

containing observations of 5G-MNCs as $\Omega = \{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_t\}$, $\mathcal{X}_t = [X_{t,1}, X_{t,2}, \dots, X_{t,B}]$, $X_{t,i} \in R^{N \times T}$ (e.g. $X_{t,1} = [x_{t,1}^1, x_{t,1}^2, \dots, x_{t,1}^T]$), where $\mathcal{X}_t$ covers one time period, such as an hour, a day, or a week. $\mathcal{X}_t$ is divided into B segments (denoted as $X_{t,i}$) and each segment occupies T samples (denoted as $x_{t,i}^T$). N is the amount of the observed base stations. The objective of this task is to design a learning framework to learn discriminative representations for distinguishing normal and anomalous behaviors, thus achieving anomaly detection in spatio-temporal data.

B. Data Augmentation

Generally, contrastive learning frameworks aim to maximize the similarity between positive pairs, which are samples and their augmentations. Therefore, applying appropriate data augmentation is crucial in contrastive learning. Considering that the KPI time series generated by 5G-MNCs typically exhibit periodic patterns that occur regularly, we propose a specific data augmentation method that treats data from the same time period as positive samples and data from different time periods as negative samples. The motivation behind this method is that the patterns during the time period from 9:00 to 10:00 differs from the one from 23:00 to 24:00, but which is similar to the patterns during the identical time period on the previous day. Because people's behaviors may vary across different time periods. The abstract framework of the designed data augmentation method is illustrated in Fig. 3.

Specifically, given a batch of data, and the batch covers only one period, which can be an hour, a day, or a week. Here, the batch covers one day. For each batch, its previous M batches are considered as its candidate batches. It is worth noting that there are also slight differences between the same time in different cycles. For example, people's behaviors on Monday morning may differ from those on Sunday morning.

To alleviate this phenomenon, multivariate dynamic time warping (MDTW) [35] is applied to select the top augmented batches that are most similar to the current batch from the M candidates. Specifically, the augmentation set selected for $\mathcal{X}_t$ is as follows:

$$dist_{t,j} = MDTW(\mathcal{X}_t, \mathcal{X}_j), j = 1, 2, \dots, M \quad (1)$$

$$\varphi_t^{aug} = \{\mathcal{X}_k \mid k \in \arg \text{sort}(dist_t)[:, K]\} \quad (2)$$

where $dist_t$ represents the MDTW distance of batch $\mathcal{X}_t$ with the other M candidate batches, and φ_t^{aug} denotes the augmentation set for $\mathcal{X}_t$ with K augmentation batches. The samples with the same index from the current batch and its augmentation batches are considered as positive samples, while those with different indexes as negative samples. To be specific, for $X_{t,i}$ in the current batch, its positive sample in the k -th augmentation segment is $X_{k,i}$, and for all $X_{k,j}$ ($j \neq i$) are its negative samples.

The data collected in real scenes is often affected by noise and external interference. To improve the robustness of detection, the generated positive and negative samples need to have generality. In this article, intuitionistic fuzzy set is introduced into data augmentation, based on which MDTW distances are calculated [39], [40]. Specifically, given a segment of data $X_{t,i}$, it passes through a fuzzy operator to generate the output data. First, the membership degree is calculated as follows:

$$\mathcal{U}(X_{t,i}) = e^{\left(\frac{X_{t,i} - \mu_{t,i}}{\sigma_{t,i} + \xi}\right)^2} \quad (3)$$

where $\mu_{t,i}$ is the mean of $X_{t,i}$, $\sigma_{t,i}$ is the standard deviation of $X_{t,i}$, ξ belongs a very small decimal set to prevent zero in the denominator.

Considering the uncertainty of membership degrees, hesitation [41] indicates a greater degree of uncertainty. To adapt to fluctuations in sequence data, refer to [41], the hesitation degree can be obtained online based on Femi formula as follows:

$$\mathcal{V}(X_{t,i}) = \frac{1}{1 + \exp(u(X_{t,i}))} + \epsilon u(X_{t,i}) \quad (4)$$

where ϵ is the interference value, which is set to 0.25 based on discussion in the experimental section.

As a result, each segment is converted into the intuitionistic fuzzy set that encompasses both membership degrees and hesitations as follows: $X'_{t,i} = [\mathcal{U}(X_{t,i}), \mathcal{V}(X_{t,i})]$, $\mathcal{X}'_t = \{[\mathcal{U}(X_{t,1}), \mathcal{V}(X_{t,1})], \dots, [\mathcal{U}(X_{t,B}), \mathcal{V}(X_{t,B})]\}$.

Thus, MDTW distance is to calculate the distance between two intuitionistic fuzzy sets. Here, three distance methods are applied, i.e. the Euclidean distance, Hamming distance and the cosine similarity. The calculation method of Euclidean distance between two intuitionistic fuzzy sets is as follows:

$$\begin{aligned} d(\mathcal{X}'_t, \mathcal{X}'_j) &= d\left(\{[\mathcal{U}(X_{t,i}), \mathcal{V}(X_{t,i})], \dots, [\mathcal{U}(X_{t,B}), \mathcal{V}(X_{t,B})]\}, \{[\mathcal{U}(X_{j,i}), \mathcal{V}(X_{j,i})], \dots, [\mathcal{U}(X_{j,B}), \mathcal{V}(X_{j,B})]\}\right) \\ &= \left\{ \frac{1}{B} \sum_{i=1}^B \left\{ \left| \mathcal{U}_{\mathcal{X}'_t}(X_{t,i}) - \mathcal{U}_{\mathcal{X}'_j}(X_{j,i}) \right|^2 \right. \right. \end{aligned}$$

$$\left. + \left| \mathcal{V}_{\mathcal{X}'_t}(X_{t,i}) - \mathcal{V}_{\mathcal{X}'_j}(X_{j,i}) \right|^2 \right\}^{\frac{1}{2}} \quad (5)$$

The distance between $\mathcal{X}'_t$ and $\mathcal{X}'_j$ is $d(\mathcal{X}'_t, \mathcal{X}'_j)$. And it satisfies the following properties:

- (1) $0 \leq d(\mathcal{X}'_t, \mathcal{X}'_j) \leq 1$;
- (2) $d(\mathcal{X}'_t, \mathcal{X}'_j) = d(\mathcal{X}'_j, \mathcal{X}'_t)$;
- (3) $d(\mathcal{X}'_t, \mathcal{X}'_j) = 0$, if and only if $\mathcal{X}'_t = \mathcal{X}'_j$.

The Hamming distance formula and the cosine similarity function are also explored to compute the distance between two fuzzy sets for measuring the closeness of their relationship. The Hamming distance formula is as follows:

$$\begin{aligned} d(\mathcal{X}'_t, \mathcal{X}'_j) &= d\left(\{[\mathcal{U}(X_{t,i}), \mathcal{V}(X_{t,i})], \dots, [\mathcal{U}(X_{t,B}), \mathcal{V}(X_{t,B})]\}, \{[\mathcal{U}(X_{j,i}), \mathcal{V}(X_{j,i})], \dots, [\mathcal{U}(X_{j,B}), \mathcal{V}(X_{j,B})]\}\right) \\ &= \frac{1}{B} \sum_{i=1}^B \left\{ \left| \mathcal{U}_{\mathcal{X}'_t}(X_{t,i}) - \mathcal{U}_{\mathcal{X}'_j}(X_{j,i}) \right| \right. \\ &\quad \left. + \left| \mathcal{V}_{\mathcal{X}'_t}(X_{t,i}) - \mathcal{V}_{\mathcal{X}'_j}(X_{j,i}) \right| \right\} \quad (6) \end{aligned}$$

And the cosine similarity function is as follows (7), as shown at the bottom of the next page.

In the experimental section, we conducted an investigation of three different calculation methods. The experiment proved that utilizing the Euclidean distance for calculating the distance between two intuitionistic fuzzy sets works better.

C. Multi-Scale Contrastive Modules

Spatio-temporal data typically exhibit intrinsic correlations in both temporal and spatial dimensions. Consequently, data with spatio-temporal characteristics often display interdependence and mutual influence. From the perspective of spatial features, taking 5G-MNC data as an example, one can naturally obtain a graph structure as shown in Fig.1. Therefore, given that graph neural networks have demonstrated strong capabilities in processing graph-structured data, we choose to convert spatio-temporal data into graph structures and utilize graph learning methods for anomaly detection. Furthermore, the dependencies within spatio-temporal data are typically nonlinear and exhibit different characteristics across various temporal and spatial scales. Therefore, single-scale analysis methods are often insufficient for comprehensively revealing anomaly patterns in the data. To capture the comprehensive anomaly information present in spatio-temporal data, a multi-scale spatio-temporal contrastive module is designed. This module enhances the ability to detect anomalies by learning consistency information at different scales. Based on this intuition, we develop a multi-scale spatio-temporal contrastive module that integrates temporal and spatial features to capture more anomaly information.

The multi-scale spatio-temporal contrastive module includes two main components: the multi-scale module and the Graph Attention Network (GAT) module. In the multi-scale module, it mainly includes a node-subgraph contrastive network and a subgraph-subgraph contrastive network.

In the node-subgraph contrastive network, the primary objective is to capture anomalous information by comparing

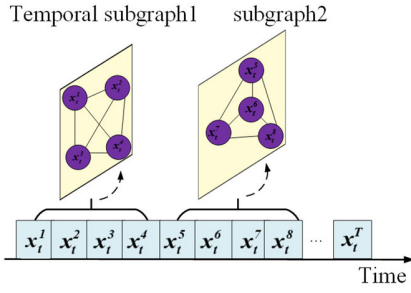


Fig. 4. An example of the temporal subgraph construction process.

the consistency between node information and its neighboring information. Node-level anomalies can cause their neighboring nodes to exhibit abnormal patterns, meaning that local anomalies can also trigger global anomalies. By learning the consistency of local and global information, the model can capture potential anomalous features. Specifically, the target node and the corresponding subgraph need to be mapped into the same embedding space. For subgraph embeddings, node features are propagated and aggregated through GCN, followed by obtaining the final embedding representation via an average readout function. The target node is mapped to the same embedding space as the subgraph through a Multi-layer Perceptron (MLP). To measure the consistency between subgraph embeddings and target node embeddings, a bilinear function is used for calculation. The subgraph-subgraph contrastive network has a similar structure to the node-subgraph network. It primarily captures anomalies by comparing different subgraphs, identifying differences and irregularities in the data.

For example, consider base stations that record network performance indicators (KPIs) at different time points. These base stations can be taken as nodes in the graph, with edges between nodes representing their geographical proximity. To detect anomalies, we first generate subgraphs from different time segments. The node-subgraph contrastive network learns the features of each base station in different time segments and checks whether these features are consistent with those of its neighboring base stations. The subgraph-subgraph contrastive network captures anomalous information between entire subgraphs. When cross effects of multiple base stations within a certain time segment exhibit abnormal patterns, that time segment might have a global anomaly. For the segment $X_{t,1} = [x_t^1, x_t^2, \dots, x_t^T]$, $x_t^1 \in R^{N \times 1}$ the procedure of constructing temporal subgraphs is shown in Fig. 4.

Before entering the multi-scale module, two parallel 1D convolution operations are initially applied to extract high-level features from both temporal and spatial perspectives. Typically, given an input sample $X^i \in R^{N \times T}$, the input to the spatial encoder E_{spat} is X_{spat}^i and the input to

the temporal encoder E_{temp} is X_{temp}^i . The feature extraction formula is as follows:

$$F_{spat}^i = \text{conv}(X_{spat}^i) \quad (8)$$

$$F_{temp}^i = \text{conv}(X_{temp}^i) \quad (9)$$

where F_{spat}^i and F_{temp}^i are the high-level features of the spatial and temporal views of the augmentation data, respectively.

1) *Node-Subgraph Contrastive Network*: Specifically, in the node-subgraph contrastive network, the subgraphs are used as inputs, and graph convolutional networks (GCN) are employed for learning. GCN is a deep learning model based on graph structures, which allows for the propagation and aggregation of node features on graphs. First, the ℓ -th node embedding in the subgraph obtained through GCN is as follows:

$$\begin{aligned} H_{spat}^{\ell_s} &= GCN \left(\widetilde{A_{spat}^{\ell_s}} M_{spat}^{\ell_s} W_{spat} \right) \\ &= \sigma \left(\widetilde{D_{spat}^{\ell_s}}^{-\frac{1}{2}} \widetilde{A_{spat}^{\ell_s}} \widetilde{D_{spat}^{\ell_s}}^{-\frac{1}{2}} M_{spat}^{\ell_s} W_{spat} \right) \end{aligned} \quad (10)$$

$$\begin{aligned} H_{temp}^{\ell_t} &= GCN \left(\widetilde{A_{temp}^{\ell_t}} M_{temp}^{\ell_t} W_{temp} \right) \\ &= \sigma \left(\widetilde{D_{temp}^{\ell_t}}^{-\frac{1}{2}} \widetilde{A_{temp}^{\ell_t}} \widetilde{D_{temp}^{\ell_t}}^{-\frac{1}{2}} M_{temp}^{\ell_t} W_{temp} \right) \end{aligned} \quad (11)$$

where ℓ_s and ℓ_t are the ℓ -th node in the spatial and temporal subgraphs respectively. $M_{spat}^{\ell_s}$ is the ℓ -th node feature matrix in the spatial subgraph. $\widetilde{A_{spat}^{\ell_s}} = (A_{spat}^{\ell_s} + I)$ is the adjacency matrix added self-loop. $\widetilde{D_{spat}^{\ell_s}}$ is the degree matrix. W_{spat} is the weight matrix of the GCN layer. $\sigma(\cdot)$ is the activation function, here the ReLU activation function is used.

Then, $H_{spat}^{\ell_s}$ and $H_{temp}^{\ell_t}$ are mapped to an embedding space using an embedding function $f(\cdot)$, resulting in embedding representations. The embedding representations are obtained as:

$$Z_{spat}^{\ell_s} = f(H_{spat}^{\ell_s}) \quad (12)$$

$$Z_{temp}^{\ell_t} = f(H_{temp}^{\ell_t}) \quad (13)$$

The final representations of the subgraphs are then computed by a Readout function, and here we use the average function.

$$Z_{spat}^{sub} = \text{AvgReadout} \left(Z_{spat}^{\ell_s} \right) = \sum_{\ell_s=1}^{m_{spat}^{sub}} \frac{Z_{spat}^{\ell_s}}{m_{spat}^{sub}} \quad (14)$$

$$Z_{temp}^{sub} = \text{AvgReadout} \left(Z_{temp}^{\ell_t} \right) = \sum_{\ell_t=1}^{m_{temp}^{sub}} \frac{Z_{temp}^{\ell_t}}{m_{temp}^{sub}} \quad (15)$$

$$\begin{aligned} \text{cosin} &= \text{Dot}(x'_t, x'_j) \\ &= \frac{\sum_{i=1}^B \{ [u(x_{t,i}) \cdot u(x_{j,i})] + [v(x_{t,i}) \cdot v(x_{j,i})] \}}{\sqrt{\sum_{i=1}^B \{ [u(x_{t,i})]^2 + [v(x_{t,i})]^2 \}} \times \sqrt{\sum_{i=1}^B \{ [u(x_{j,i})]^2 + [v(x_{j,i})]^2 \}}} \end{aligned} \quad (7)$$

where m_{spat}^{sub} and m_{temp}^{sub} are the numbers of nodes in the spatial and temporal subgraphs respectively.

For the target node, we use multilayer perceptron (MLP) to map it into the same embedding space as the subgraph. The time step node embedding is similarly to the target node. The feature matrix of the target node is denoted as M_{spat}^{node} . The calculation process of MLP is as follows:

$$Z_{spat}^{node} = MLP(M_{spat}^{node} W_{spat}) = \sigma(M_{spat}^{node} W_{spat}) \quad (16)$$

$$Z_{temp}^{node} = MLP(M_{temp}^{node} W_{temp}) = \sigma(M_{temp}^{node} W_{temp}) \quad (17)$$

Here, the weight is shared with the GCN.

To measure the consistency between the subgraph embedding and the target node embedding, a bilinear function is utilized to calculate the similarity between them.

$$S_{spat}^{n-s} = Bilinear(Z_{spat}^{sub}, Z_{spat}^{node}) = Z_{spat}^{sub} W_{Biliner} Z_{spat}^{node} \quad (18)$$

$$S_{temp}^{n-s} = Bilinear(Z_{temp}^{sub}, Z_{temp}^{node}) = Z_{temp}^{sub} W_{Biliner} Z_{temp}^{node} \quad (19)$$

where $W_{Biliner}$ is the weight matrix of the bilinear function.

2) *Subgraph-Subgraph Contrastive Network*: The subgraph-subgraph contrastive network is similar in structure to the node-subgraph contrastive network. The final embedding representations of the subgraphs obtained in the subgraph-subgraph contrastive network are as follows:

$$Z_{spat}^{sub} = AvgReadout(Z_{spat}^{\ell_s}) = \sum_{\ell_s=1}^{m_{spat}^{sub}} \frac{Z_{spat}^{\ell_s}}{m_{spat}^{sub}} \quad (20)$$

$$Z_{spat}^{sub'} = AvgReadout(Z_{spat}^{\ell_s'}) = \sum_{\ell_s=1}^{m_{spat}^{sub'}} \frac{Z_{spat}^{\ell_s'}}{m_{spat}^{sub'}} \quad (21)$$

$$Z_{temp}^{sub} = AvgReadout(Z_{temp}^{\ell_t}) = \sum_{\ell_t=1}^{m_{temp}^{sub}} \frac{Z_{temp}^{\ell_t}}{m_{temp}^{sub}} \quad (22)$$

$$Z_{temp}^{sub'} = AvgReadout(Z_{temp}^{\ell_t'}) = \sum_{\ell_t=1}^{m_{temp}^{sub'}} \frac{Z_{temp}^{\ell_t'}}{m_{temp}^{sub'}} \quad (23)$$

Here, we use $Z_{spat}^{sub'}$ to denote the embedding representation of another subgraph and $m_{spat}^{sub'}$ to denote the number of nodes in another subgraph.

Next, the same bilinear function is used to compute their similarity.

$$S_{spat}^{s-s} = Bilinear(Z_{spat}^{sub}, Z_{spat}^{sub'}) = Z_{spat}^{sub} W'_{Biliner} Z_{spat}^{sub'} \quad (24)$$

$$S_{temp}^{s-s} = Bilinear(Z_{temp}^{sub}, Z_{temp}^{sub'}) = Z_{temp}^{sub} W'_{Biliner} Z_{temp}^{sub'} \quad (25)$$

Finally, the results of the node-subgraph contrastive network and the subgraph-subgraph contrastive network are concatenated using a concatenation operation to serve as the inputs for the next step.

$$S_{spat}^i = concat(S_{spat}^{n-s}, S_{spat}^{s-s}) \quad (26)$$

$$S_{temp}^i = concat(S_{temp}^{n-s}, S_{temp}^{s-s}) \quad (27)$$

3) *Advanced Feature Extraction*: To fully utilize the spatial and temporal structure information of the graph network to detect anomalous behaviors, two parallel GAT layers with multi-head attention mechanisms are used to capture the relationships between nodes. For a general graph with n nodes, i.e., $\{h_1, h_2, \dots, h_n\}$, adj is the adjacency matrix constructed from the nodes. The output of the GAT layer is as follows:

$$G_{spat}^i = GatConv(S_{spat}^i, adj) \quad (28)$$

$$G_{temp}^i = GatConv(S_{temp}^i, adj) \quad (29)$$

After that, a global maximum pooling operation is used to extract the key features.

$$P_{spat}^i = GlobalMaxPool(G_{spat}^i) \quad (30)$$

$$P_{temp}^i = GlobalMaxPool(G_{temp}^i) \quad (31)$$

The fully connected layer is then used to obtain the most representative feature representations.

$$r_{spat}^i = Dense(P_{spat}^i) \quad (32)$$

$$r_{temp}^i = Dense(P_{temp}^i) \quad (33)$$

Finally, a project layer f_{proj} is used to obtain a fused representation:

$$Z_i = f_{proj}(r_{spat}^i \oplus r_{temp}^i) \quad (34)$$

The obtained fusion results are called time-slice embeddings (TSEs) because they represent the embedded potential representations in a certain time slice.

4) *Loss Function*: To ensure that time slice embeddings (TSEs) from different time slices are distinguishable in potential space, a customized loss of contrast is designed. TSEs with the same indices from the current batch and its augmentation batch are considered positive samples, while TSEs with different indices are considered negative samples. Let $\mathcal{X}_t$ denote the current batch and $\mathcal{X}_k$ denote its k -th augmentation batch. Then $X_{t,i} \in \mathcal{X}_t$ and $X_{k,i} \in \mathcal{X}_k$ are form a positive input sample pair, and $z_{t,i}$ and $z'_{k,i}$ are the corresponding learned TSEs. The loss of contrast between the current batch and its k -th augmented batch can be expressed as:

$$L_k = - \sum_{i=1}^B \log \frac{\exp\left(\text{sim}\left(z_{t,i}, z'_{k,i}\right) / \tau\right)}{\sum_{j=1}^B I_{[i \neq j]} \exp\left(\text{sim}\left(z_{t,i}, z'_{k,j}\right) / \tau\right)} \quad (35)$$

where $I_{[i \neq j] \in \{0,1\}}$ is an indicator function and is 1 when $i \neq j$, 0 otherwise. τ is a temperature parameter. The total loss is the weighted sum of the current batch and its corresponding K augmented batches, as follows:

$$L = \sum_{k=1}^K \frac{\beta_k}{B} L_k \quad (36)$$

where β_k is the coefficient between the current batch and its k -th augmentation batch, and its formula is:

$$\beta_k = \frac{dist_{t,k}}{\sum_{u=1}^K dist_{t,u}} \quad (37)$$

The training procedure of the proposed method is summarized in Algorithm 1.

Algorithm 1 Training Algorithm of the Proposed Method

Input: Training set $\Omega = \{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_t\}$, where $\mathcal{X}_t = [X_{t,1}, X_{t,2}, \dots, X_{t,B}]$.
 M : The number of candidate batches.
 K : The number of augmented batches.
Output: A trained MFCAD model with parameters sets Θ .

for each iteration **do**
 for current batch $\mathcal{X}_t$ in Ω **do**
 Step1: Select M candidate batches for $\mathcal{X}_t$.
 Step2: For each $X_{t,i}$ in $\mathcal{X}_t$ fuzzy the data into an intuitive fuzzy set containing membership degrees and hesitation degrees:
 $X'_t = \{[\mathcal{U}(X_{t,1}), \mathcal{V}(X_{t,1})], \dots, [\mathcal{U}(X_{t,B}), \mathcal{V}(X_{t,B})]\}$.
 Step3: Calculate the MDTW distance between current batch and its candidate augmented batches to obtain the top- K candidates so as to form augmentation set φ_t^{aug} .
 Step4: Feed $\mathcal{X}_t$ and its augmented batches from φ_t^{aug} into the multi-scale spatio-temporal contrastive model to obtain the generated TSEs for forming sample pairs.
 Step5: Calculate the batch loss using Eq. (36).
 Step6: Use optimizer to train the model.
 end for
end for

D. Anomaly Detection

To achieve anomaly detection in latent space, a clustering method is applied for each group of TSEs, where the clustering centers are regarded as the corresponding prototypes. To this end, the FCM clustering method is utilized. Given a group of TSEs $Z_t = \{z_{t,1}, z_{t,2}, \dots, z_{t,B}\}$, where $z_{t,i}$ denotes the i -th learned TSE from the t -th group. Calculating the clustering centers can be achieved by minimizing the following problem.

$$\begin{aligned} \min J &= \sum_{r=1}^C \sum_{i=1}^B u_{rp}^m (z_{t,i} - c_{t,r})^2 \\ \text{s.t. } &\sum_{r=1}^C u_{ri}^m = 1, \quad \forall i = 1, 2, \dots, B \end{aligned} \quad (38)$$

where C is the number of centers for a group, $c_{t,r}$ is the r -th center for the t -th TSE group, u_{ri}^m is the membership degree of sample i belonging to the r -th center, and m is a weighted index. The ℓ_2 distance between the TSE and its corresponding prototypes is used to measure the inference score:

$$s_{t,i} = \min \{\|z_{t,i} - c_{t,1}\|_2, \|z_{t,i} - c_{t,2}\|_2, \dots, \|z_{t,i} - c_{t,C}\|_2\} \quad (39)$$

where $s_{t,i}$ is the inference score for the i -th TSE which calculates the minimum ℓ_2 distance between the TSE and its corresponding prototypes. An input sample is considered anomalous if its TSE is distant from all the corresponding prototypes. Specifically, we identify an anomaly if the inference score is larger than a threshold, where the Drift Peak Over Threshold (DPOT) [44] is adopted to choose the threshold dynamically.

IV. EXPERIMENTS

In this section, the datasets and experimental settings are introduced. Subsequently, the performances of MFCAD are assessed using real-world 5G-MNC data and public datasets.

An interpretation analysis is then presented, followed by conducting ablation studies and sensitivity analyses. Additionally, ablation studies and sensitivity analyses are carried out. Lastly, the model's generalizability is validated by applying it to an industrial dataset.

A. 5G-MNC Dataset Description

To evaluate the effectiveness of the proposed method, the real-world 5G-MNC datasets are used. The datasets are collected by China Mobile Communication, and they contain the KPI records from two 5G-MNCs. Wherein, MNC 1 contains 11 base stations, and MNC 2 contains 4 base stations. The selected KPI for performing anomaly detection is the number of Radio Resource Control (RRC) connection establishment requests, and the KPI data is aggregated every 15 minutes. The time period is from 3rd Sep 2021 to 3rd Jan 2022, and the first two months of historical records are used as a training set and validation set, and the rest serves as a test set. The statistics of the two MNCs are shown in Table I.

In practical scenarios, anomalies within a 5G-MNC generally manifest in a continuous manner. Consequently, for two datasets, the anomalies are annotated segment-wise, i.e., if a segment contains an anomalous pattern, the entire segment is annotated as anomalous. The annotation of anomaly labels is carried out by domain experts from China Mobile. Two illustrative examples are provided in Fig. 5. In the green area of Fig. 5(a), there is an obvious boost in a base station. However, sudden changes in a certain base station do not mean anomalies in the whole network system. While in Fig. 5(b), one can find that the areas marked in red demonstrate anomalous patterns compared to the other areas, which indicates there are anomalies occurring in the system.

B. Experimental Settings

1) *Evaluation Metric:* To measure and evaluate the performance of the method, three common metrics, precision, recall, and F1-score are utilized. These metrics are derived from the following expressions:

$$Precision = \frac{TP}{TP + FP} \quad (40)$$

$$Recall = \frac{TP}{TP + FN} \quad (41)$$

$$F1 - score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (42)$$

where TP is the true positives, FP represents the false positives, and FN stands for the false negatives. In real applications, the anomalies generally occur in a continuous manner, thus we do not care about the point-wise metrics. Following the same evaluation strategy as [17], we mark the whole segment as positive if any anomalous observation is in this segment.

2) *Baselines:* We compare the proposed method with state-of-the-art anomaly detection methods, which are AE [45], VAE [46], Local Outlier Factor (LOF) [47], Isolation Forest (IF) [48], LSTM-VAE [49], DAGMM [20], MGCLAD [15], and DCdetector [16]. Specifically, all the baselines follow the

TABLE I
DATASET STATISTICS

	MNC 1	MNC 2
Total length	11808	11808
Number of sequences	11	4
Number of test segments	720	720
Number of anomaly segments	34	19
Anomaly rate (%)	4.72%	2.63%

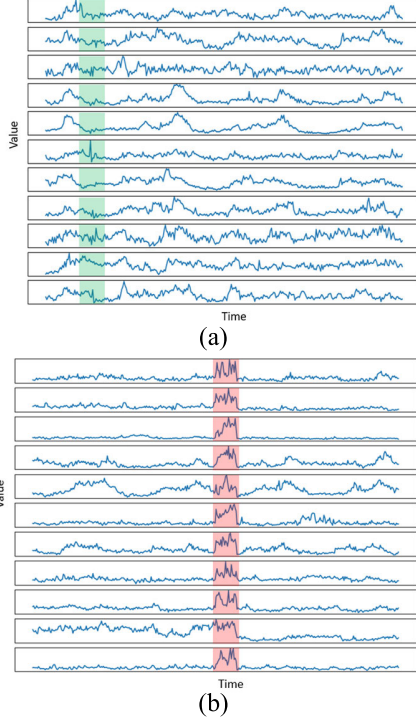


Fig. 5. Examples of 5G-MNC data.

architecture proposed in the original articles. Since not all the anomaly detection methods are designed for 5G-MNC anomaly detection, we performed all possible thresholds for every model and reported the experimental results according to the highest F1-score.

3) *Implementation Details*: The proposed method is implemented in Tensorflow Library and is trained on a 64-bit Intel core i7-6700CPU@ 3.40GHz/16GB RAM configuration computer. The Adam optimizer with learning rate is used for training, and the training epochs are set to 50. The time slice length T is set to 8, which means each time slice contains two hours for sampling.

C. Data Preprocessing

Generally, there are often missing values and huge spike values involved in the original data. Since the ratio of these values is less than 5% in each series, just as in [50], we simply remove the data that deviates the most from the mean value and replace them with their neighboring normal observations. Besides, we perform data normalization on both training and testing sets to improve the robustness of the proposed method. Furthermore, same as [51], the adjacency matrix of a 5G-MNC

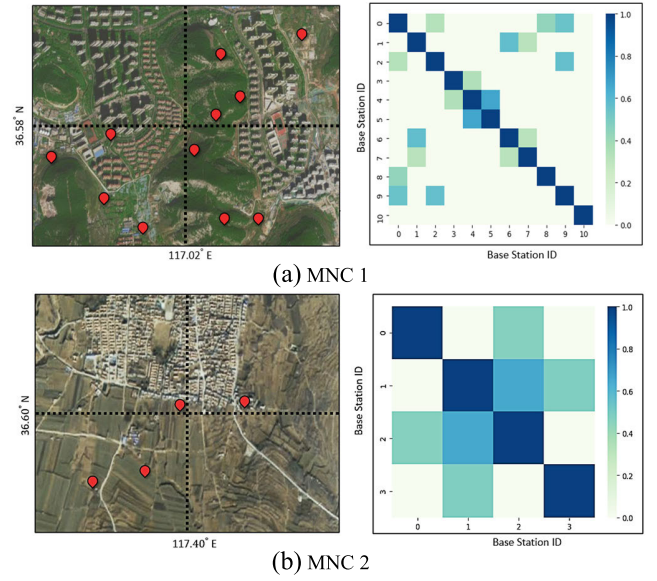


Fig. 6. Two clusters of 5G-MNCs (each marker denotes a base station) and corresponding heat maps of weighted adjacency matrices.

is computed based on the distances among base stations in the mobile networked cluster. The weighted adjacency matrix can be formed as:

$$a_{ij} = \begin{cases} \exp\left(-\frac{\eta_{ij}^2}{\sigma^2}\right), & i \neq j \text{ and } \exp\left(-\frac{\eta_{ij}^2}{\sigma^2}\right) \geq \delta \\ 0, & \text{otherwise.} \end{cases} \quad (43)$$

where a_{ij} is the weight of the edge, and it is determined by η_{ij} , which is the distance between base station i and j . σ^2 and δ are the thresholds to control the distribution and sparsity of adjacency matrix, and they are set to $1e-5$ and 0.3 in this work. The two selected 5G-MNCs and corresponding heat maps of weighted adjacency matrixes are shown as Fig. 6.

D. Parameter Comparison

During the experiment, we found that setting a suitable interference value when calculating the hesitation degree can better reflect the uncertainty. By setting different interference values for exploration, the experimental results show that the model's performance is best when the interference value is set to 0.25. That is to say, the experimental performance metrics, including recall, precision, and f1 score, are relatively high when the interference value is set to 0.25. This can be visually presented in Fig. 7.

An appropriate distance formula is needed to measure the similarity between two fuzzy sets when calculating the MDTW distance. In this paper, we explore three distance formulas including Euclidean distance, Hamming distance, and cosine similarity function. The experimental results demonstrate that using Euclidean distance yields the best performance compared to the Hamming distance and cosine similarity function. This can be observed in Fig. 8.

E. Experimental Results

In this subsection, we conduct experiments on real-world 5G-MNC datasets to evaluate the effectiveness of MFCAD.

TABLE II
PERFORMANCE OF WADI AND SWAT DATASETS ON DIFFERENT BASELINES

Method	MNC 1			MNC 2		
	Recall	Precision	F1-score	Recall	Precision	F1-score
AE	0.862	0.982	0.918	0.884	0.976	0.927
IF	0.795	0.994	0.883	0.815	0.911	0.860
LOF	0.755	0.956	0.843	0.833	0.927	0.877
VAE	0.798	0.964	0.873	0.934	0.981	0.957
LSTM-VAE	0.943	0.965	0.953	0.922	0.996	0.958
DAGMM	0.959	0.962	0.961	0.954	0.978	0.966
MGCLAD	0.963	0.891	0.926	0.999	0.934	0.965
DCdetector	0.941	0.985	0.962	0.986	0.982	0.981
MFCAD	0.985	0.984	0.985	0.990	0.988	0.989

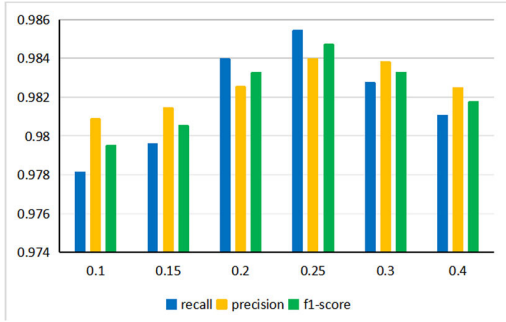


Fig. 7. Experimental effects at different interference values.

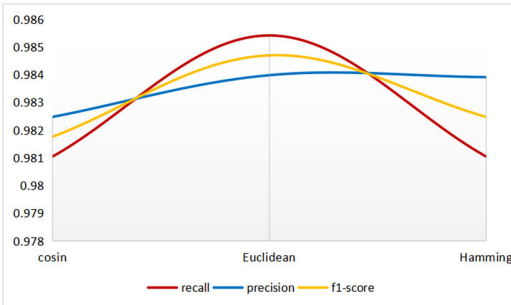


Fig. 8. Experimental effects of using different distance formulas.

The overall experimental results of MFCAD with six methods are demonstrated in Table II, where the best F1-score achieved is in bold.

IF and LOF achieve the lowest F1-score since they are both unsupervised anomaly detection methods that do not capture temporal and spatial information. However, the information hidden in the raw data is critical for anomaly detection. AE, VAE, and LSTM-VAE are reconstruction-based anomaly detection methods, which use sequential data as input so that the temporal information can be retained. Nonetheless, reconstruction-based anomaly detection methods mainly focus on the performance of sample reconstruction, and do not care about the distinguishability of the encoded features for abnormal and normal samples. Thus, it would cause even abnormal data to be well reconstructed. Moreover, the prediction-based

anomaly detection method, like DAGMM, tries to predict the values based on historical data and performs anomaly detection according to the forecasting errors. However, some data are inherently unpredictable due to various internal/external factors, especially complex, dynamic mobile network data. Therefore, normal observations would be easily mistakenly detected by DAGMM and cause a false positive. In the proposed MFCAD, the spatio-temporal patterns of the original data at different time intervals in a cycle are fully explored. Moreover, MFCAD detects anomalies in latent space, which does not require reconstruction or prediction of features. This further mitigates the potential errors introduced by the model itself. Therefore, MFCAD achieves the best performance compared with other methods.

F. Visualized Explanation

To explain the performance of the proposed method on 5G-MNC data, the TSEs from the same time slice generated by MNC 1 in different cycles are visualized in Fig. 9. From the figure, one can see that most TSEs are gathered together indicating normal patterns, while the abnormal TSEs are farther apart. To validate this phenomenon, we show the original data corresponding to several of these TSEs. Upon observation, one can see that the TSEs of A and E behave abnormally, while the others behave normally.

Furthermore, to demonstrate the relationships between different TSEs, we also project all the learned TSEs into 2D embedding space for visualization. The t-SNE plot is shown in Fig. 10 where each colored point denotes an embedding vector of TSE from a different time slice and different colored star symbols represent the corresponding prototypes for each group of TSEs. From the figure, one can see that the TSEs have a significant aggregation phenomenon. It indicates that the TSEs from the same time slices share similar patterns, while the TSEs from different time slices share distinct patterns.

G. Ablation Study

In this subsection, the MNC 1 dataset is used for analyzing the influence of different modules in the model. Detailly,

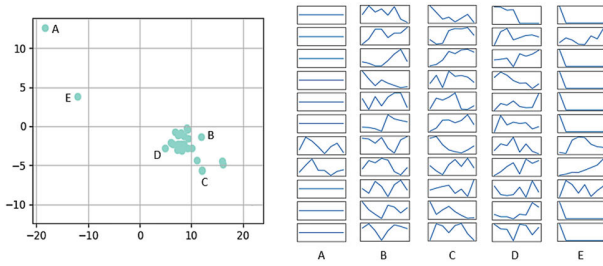


Fig. 9. A t-SNE plot of the TSEs from same time slice and the time patterns corresponding to five of these TSEs.

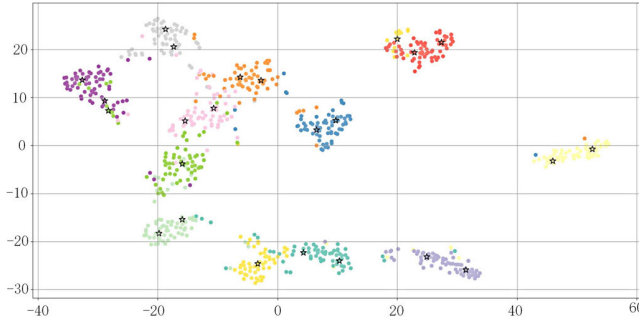


Fig. 10. A t-SNE plot of different TSEs. Each color represents a different time slice of a cycle, and the star symbols indicate the prototypes of each group of TSEs.

TABLE III

COMPARISON WITH DIFFERENT VARIANTS

Method	Recall	Precision	F1-score
w/o spatial	0.973	0.983	0.978
w/o temporal	0.976	0.981	0.978
w/o data augmentation	0.954	0.939	0.946
w/o clustering	0.977	0.900	0.937
w/o multi-scale	0.954	0.913	0.933
MFCAD	0.985	0.984	0.985

TABLE IV

PERFORMANCE OF THE PROPOSED MODEL AND BASELINES

Datasets	Features	Train	Test	Anomalies
SWaT	51	495,000	449,919	12.14%
WADI	127	789,371	172,801	5.77%

we perform an ablation study by disabling a certain module of MFCAD. Wherein, w/o spatial denotes the variant without spatial GAT layer, and w/o temporal denotes the variant without temporal GAT layer. We also compare MFCAD with the variant that simply uses the last consecutive cycle as an augmentation set and it is denoted by w/o data augmentation. The variant denoted by w/o clustering performs anomaly detection by using the distance between the two TSEs of adjacent cycles to analyze the effect of the cluster module. w/o multi-scale denotes the variant without multi-scale contrastive module. The results are shown in Table III, from which one can see that all the above modules of MFCAD are indispensable.

The variant without temporal or spatial module extracts features from only one view, which prevents the full exploration of the spatial and temporal correlation of the data.

The designed data augmentation method enables a robust contextual representation for each time slice in a cycle, therefore, a performance decrease is observed after disabling data augmentation. Performing anomaly detection by the distance between two TSEs of adjacent cycles uses only local information, while global information cannot be utilized, thus the variant without clustering performs poorly. In conclusion, the ablation of any of the modules in the designed architecture leads to poorer performance. The variant without a multi-scale contrastive module solely captures anomalies from a single scale. This restricts its capacity to capture anomalous patterns in neighboring time or space. As a result, its performance is diminished. In summary, the removal of any module within the architectural design can result in a decline in performance.

H. Sensitivity Analysis

In this subsection, experiments using the MNC 1 dataset to analyze the sensitivity of MFCAD have been done. Specifically, the dimension of TSEs D , the candidate augmentation set size M , and the top- K augmentation set size K are utilized for analysis. Fig. 11(a) shows the impact of a key hyper-parameter D . Increasing the dimension of TSEs can improve the capability of representation. However, a larger dimension can harm performance as it adds redundant information. Therefore, we empirically set the value of D to 18. Fig. 11(b) demonstrates the results of varying M and K in a range since they are used jointly to control the size of the data augmentation. According to the results, the F1 score shows an upward trend with the increase of M and K and tend to flatten out at the end. Such results demonstrate that applying an appropriate size of data augmentation can improve the performance, while excessive augmented data cannot guarantee the robustness of the learned representations and further increase the burden of training. Therefore, in practice, K is set to 3 and M is set to 7.

Furthermore, in anomaly detection of multivariate time series, the window size has an important impact on the performance of the model. Different window sizes affect the model's ability to capture anomalies and its detection accuracy. Therefore, we explore the effect of window size on model performance using the 5G-MNC datasets. The experimental results are shown in Fig. 12. The performance is relatively good when the window size is around 12. And too large or too small size will make the model performance decrease.

I. Industrial Data Applications

1) *Datasets*: To further verify the generality of the proposed method, two sensor datasets based on water treatment physical testbed systems SWaT and WADI are used for validation. The SWaT dataset is derived from a water treatment testbed coordinated by the Public Utilities Board of Singapore, which is a dataset used for cybersecurity field research based on a dataset of actual water treatment systems. The SWaT dataset consists of 11 days of continuous operation, of which 7 days of data were collected under normal operation, while 4 days of data were collected through attack scenarios. WADI is an extension of SWaT that absorbs some of the reverse

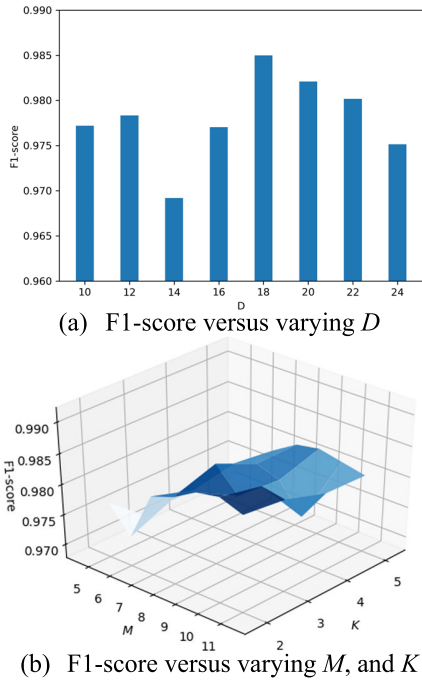


Fig. 11. Sensitive experiments on partial hyper-parameters.

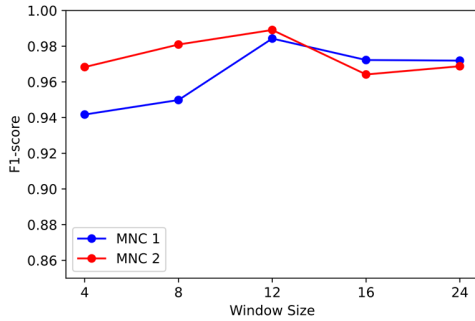


Fig. 12. The effect of window size on the performance of model.

osmosis permeate and raw water from SWaT, thereby forming a complete and authentic water treatment, storage, and distribution network. The WADI dataset consisted of 14 days of continuous normal operation and 2 days of data collection based on attack scenarios. The statistics for the two datasets are shown in Table IV. The two datasets are available at https://itrust.sutd.edu.sg/itrust-labs_datasets/.

2) Baselines:

- KNN: Categories are determined by majority vote or weighted average based on the distance between the sample to be tested and the individual samples in the training set [52].
- FB: The input signal is decomposed into a number of subbands of different frequencies and a different filter is applied to each subband for feature extraction [53].
- AE: The basic structure consists of an encoder and decoder that learns a compressed representation of the input data by minimizing the reconstruction error. Anomaly detection is performed based on the reconstruction [54].
- DAGMM: Combine a self-encoder with a Gaussian mixture model. Learning the representation of the data

through the self-encoder and then using Gaussian mixture models to model the normal data [20].

- LSTM-VAE: It Combines LSTM and variational self-encoder. By learning the temporal dependencies and latent spatial representations of the data, anomalous samples may exhibit different sequence patterns than normal samples, allowing anomaly detection by comparing the distribution of reconstruction errors or latent variables [49].
- MAD-GAN: It is a multi-class anomaly detection method based on generative adversarial networks. By training generators and discriminators, MAD-GAN can generate normal samples and distinguish abnormal samples [55].
- GDN: This approach combines graph neural networks and graph structure learning to identify anomalies by leveraging graph attention-based prediction errors [12].
- STADN: Spatio-temporal information is combined and anomaly detection is performed using GAT and Long Short-Term Memory (LSTM) networks [21].
- MGCLAD: MGCLAD utilizes a multi-view graph contrastive learning approach. It considers both the intra-signal graph structure and the relationship between different signals to improve the performance of anomaly detection [15].
- DCdetector: DCdetector is a state-of-the-art contrastive learning-based anomaly detection method. It designs a dual-branch attention structure based on contrastive learning to learn a permutation-invariant representation and guides the learning process by using contrastive loss [16].

3) *Effectiveness Assessment*: We compare the precision, recall, and F1 score of the proposed method MFCAD with the baseline, and the results are presented in Table V. Based on the experimental data, one can observe that MFCAD outperforms all baseline methods in terms of F1 scores on both datasets, showing significant improvement. F1 scores peak at 0.98 and 0.98, respectively. In addition, MFCAD improves the recall to 97.43% and 99.78%. Higher recall value indicates that the model can cover almost all anomalous samples. This proves the effectiveness of MFCAD in anomaly detection tasks, as it is able to capture the vast majority of anomalous samples. And, ablation experiments are conducted on the WADI and SWaT datasets shown in Table VI, which further verify the effectiveness of each module to the model performance.

To comprehensively evaluate the proposed method, we further incorporate AUROC and AUPRC metrics. AUROC measures the balance between the true positive rate and the false positive rate at different thresholds and is commonly used to assess the performance of classification models. The closer the value is to 1, the better the model's performance. AUPRC focuses on the balance between precision and recall, with higher values indicating better model performance. We compare the proposed method with two state-of-the-art models. As shown in the table VII, the proposed method performs well on most of the datasets.

J. Complexity and Robustness Analysis

The time complexity of the proposed method is primarily contributed by the fuzzy data augmentation method and the

TABLE V
DATA FOR TWO PUBLIC DATASETS

Method	WADI			SWaT		
	Prec(100%)	Rec(100%)	F1	Prec(100%)	Rec(100%)	F1
KNN	7.76	7.75	0.08	7.83	7.83	0.08
FB	8.60	8.60	0.09	10.17	10.17	0.10
AE	34.35	34.35	0.34	72.63	52.63	0.61
DAGMM	54.44	26.99	0.36	27.46	69.52	0.39
LSTM-VAE	87.79	14.45	0.25	96.24	59.91	0.74
MAD-GAN	41.44	33.92	0.37	98.97	63.74	0.77
GDN	97.50	40.19	0.57	99.35	68.12	0.81
STADN	98.49	45.57	0.62	99.92	70.79	0.83
MGCLAD	98.00	87.84	0.93	94.81	89.11	0.92
DCdetector	96.23	99.36	0.98	93.14	99.96	0.96
MFCAD	97.57	97.43	0.98	95.95	99.78	0.98

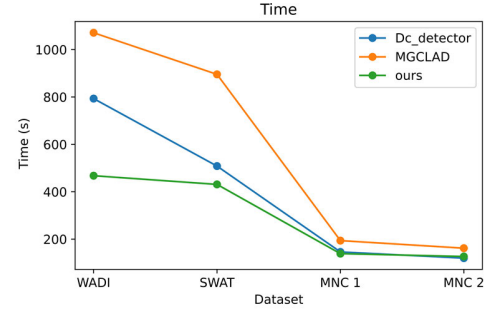
TABLE VI
RESULTS OF ABLATION EXPERIMENTS WITH WADI
AND SWAT DATASETS

Method	WADI			SWaT		
	Recall	Precision	F1-score	Recall	Precision	F1-score
w/o spatial	0.974	0.934	0.952	0.990	0.955	0.972
w/o temporal	1.000	0.932	0.965	0.992	0.955	0.973
w/o data augmentation	0.969	0.974	0.972	1.000	0.929	0.963
w/o clustering	0.989	0.900	0.942	0.983	0.796	0.879
w/o multi-scale	0.985	0.932	0.958	0.983	0.899	0.939
MFCAD	0.974	0.976	0.978	0.998	0.960	0.976

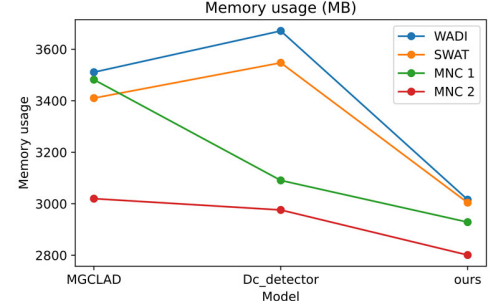
TABLE VII
AUROC AND AUPRC METRICS FOR 4 DATASETS
ON DIFFERENT MODELS

Method		MGCLAD	DCdetector	MFCAD
WADI	AUROC	0.858	0.999	0.881
	AUPRC	0.955	0.975	0.979
SWaT	AUROC	0.873	0.999	0.903
	AUPRC	0.978	0.967	0.972
MNC1	AUROC	0.905	0.862	0.929
	AUPRC	0.961	0.962	0.982
MNC2	AUROC	0.911	0.914	0.934
	AUPRC	0.940	0.962	0.974

multi-scale spatio-temporal contrastive encoder. The data augmentation method involves similarity computation and sorting, with a complexity of $O(n^2 \cdot \tilde{D} + (\frac{n}{B}) \cdot \log(\frac{n}{B}))$, where n denotes time series length, $\tilde{D}$ denotes the feature dimension, B denotes batch size. And, the time complexity of the spatio-temporal contrastive encoder is $O(n^2 \cdot D)$. Furthermore, we evaluate the efficiency of the proposed method by calculating the average running time and memory usage. The experiments run on a computer with a 64-bit Intel Core i7-6700 CPU @ 3.40GHz and 16GB RAM configuration, implemented using Python 3.6. Four datasets of different sizes are applied to fully evaluate the performance, and the datasets are described



(a) Average running time varying with different datasets



(b) Memory usages varying with different models

Fig. 13. Experiments on complexity analysis.

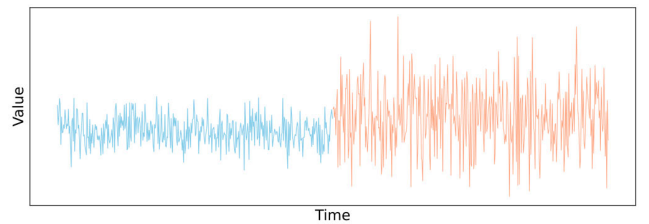


Fig. 14. An example. Light blue line represents the original time series data and the red part shows the changed data with different characteristics distribution.

in Table I and Table IV. We also compared the proposed method with two other methods based on contrastive learning. The results are shown in Fig. 13. When the data size and

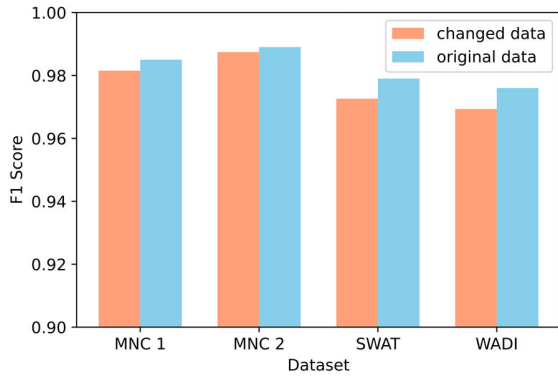


Fig. 15. Comparative experimental results between original data and changed data.

dimensionality are small (e.g., MNC1, MNC2), the difference in the running time of all methods is insignificant. When the data size and dimension are large (e.g., WADI and SWaT), the proposed method outperforms the other methods in terms of both runtime and memory consumption.

To explore the model performance under longer and evolving environment, we further introduce intervention into the datasets to simulate evolving network conditions by changing the characteristics of data distribution, as exemplified in Fig. 14. And, in four datasets, we compare the results of the proposed model between the original data and the changed data, where the experimental results are shown in Fig. 15. One can observe that data drift in characteristics distribution does not significantly degrade the performance of the proposed model, which still maintains a robust performance. These experiments indicates that the model has the capability to adapt to different environments in applications with long-term deployment.

V. CONCLUSION

We propose a novel universal anomaly detection model, namely Multi-scale Fuzzy Contrastive-based Anomaly Detection (MFCAD), which is capable of detecting anomalies in multivariate time series with spatio-temporal features. MFCAD utilizes multi-scale contrastive learning to obtain more discriminative representations and performs explicit anomaly detection in the latent space by measuring the distance between outliers and inliers. Our ablation experiments demonstrate that the spatio-temporal contrastive model, data augmentation method, and multi-scale mechanism significantly enhance the anomaly detection performance of MFCAD. Experimental results on real-world 5G-MNC datasets demonstrate that MFCAD outperforms state-of-the-art methods, achieving the best performance. Furthermore, the effectiveness of MFCAD is further validated using an industrial dataset. As a result, the proposed MFCAD model holds practical value in rapidly and accurately diagnosing anomalies.

How to effectively construct graph structure is an important issue to the performance of anomaly detection with graph learning. In this article, spatial features of data are mainly applied to construct graph structures, which is a typical and reasonable approach. However, from a data-driven perspective,

correlations existing in temporal features at different scales also have valuable information to the construction of graph structures. Therefore, in the future work, we would like to find an effective way to further integrate multi-scale temporal features on the basis of priori spatial features for graph structure construction in the application of anomaly detection.

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